Supervised Dimension Reduction And Turnout Modelling

Combined PDF version of the updated Step 1 through Step 5C reports. Tables are reformatted for readability while preserving the report contents.

Step 1: Full Supervised PLS Model

This model intentionally uses the full predictor universe before filtering obvious inverse or redundant variables. Mean turnout (outcome_mean_participation_citizen_18plus) supervises the latent components through PLS.

Model Fit

- Observations: 583
- Predictors: 70
- Selected components by 10-fold CV: 12
- Training R2: 0.693
- Cross-validated R2: 0.565
- Cross-validated RMSE: 0.057

Main Read

The full model is useful as a stress test. It lets the supervised dimension reduction procedure see every available signal, but it also lets inverse and same-concept variables double-enter the component construction. VIP scores should therefore be read as supervised importance, not as final variable-selection decisions.

The selected components are latent variables, not original predictors. In this model, 12 components means PLS built 12 turnout-supervised dimensions from the 70 original predictors. The original variables are interpreted through VIP scores, coefficients, and component loadings.

Highest VIP Variables

Variable	Family	VIP	Turnout corr	PLS direction
block3_visible_minority_share	immigration_citizenship	2.286	-0.642	lower turnout
block3_immigrant_share	immigration_citizenship	2.031	-0.573	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	1.985	0.554	higher turnout
block1_unemployment_rate_share	household_class	1.934	-0.532	lower turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.907	-0.535	lower turnout
block1_average_household_size	household_class	1.834	-0.485	lower turnout
block4_mayoral_vote_fragmentation	mayoral_competitiveness	1.775	-0.495	higher turnout
block3_non_official_mother_tongue_share	immigration_citizenship	1.751	-0.491	higher turnout
block3_non_citizen_share	immigration_citizenship	1.534	-0.417	lower turnout
block3_citizen_adult_share	immigration_citizenship	1.401	0.353	lower turnout
block1_median_age	age_structure	1.268	0.253	lower turnout
block3_recent_immigrant_share	immigration_citizenship	1.227	-0.317	lower turnout
block1_age_65_plus_share	age_structure	1.219	0.213	higher turnout
block4_mayoral_top_two_margin	mayoral_competitiveness	1.151	0.288	lower turnout
block4_mayoral_winner_margin	mayoral_competitiveness	1.151	0.288	lower turnout
block4_federal_margin	federal_competitiveness	1.065	0.272	higher turnout

Variable	Family	VIP	Turnout corr	PLS direction
block2_apartment_lt5_storeys_count	housing_form_density	1.034	0.275	higher turnout
block5_library_count_1200m	service_access	0.990	0.246	lower turnout
block3_english_french_knowledge_share	immigration_citizenship	0.969	0.266	higher turnout
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.933	0.160	higher turnout

Reference Categories Suggested By The Full Model

The full model points toward these broad reference categories for story-building: immigration and racialized geography, education and class resources, household composition, local electoral fragmentation, citizenship and language context, age structure, and urban service/contact geography. The strongest plain-language variables behind these categories are visible minority share, immigrant share, bachelor or higher education share, unemployment rate, effective mayoral candidates above five percent, average household size, mayoral vote fragmentation, non official mother tongue share, non citizen share, citizen adult share, median age, recent immigrant share.

Short Interpretation

Variables with VIP above 1 are contributing more than average to the turnout-supervised PLS projection. The strongest variables should be carried into Step 2 and Step 3 as evidence, but not accepted mechanically. For example, if renter and owner variables both score highly, that is not two separate housing-tenure findings; it is the same tenure concept appearing twice with opposite coding.

Step 2: Multicollinearity and Redundancy Diagnostics

This step diagnoses the full predictor universe used in Step 1. It does not decide the final variable set by itself. It flags where the same concept is probably being counted more than once.

Headline Findings

- Predictor count checked: 70
- High-correlation pairs with |r| >= 0.70: 51
- Near duplicate or inverse pairs with |r| >= 0.95: 10
- Variables with VIF >= 10: 43
- Variables with VIF >= 5: 51

Highest VIF Variables

Variable	Family	VIF
block2_apartment_lt5_storeys_count	housing_form_density	inf
block2_apartment_total_count	housing_form_density	inf
block5_no_car_household_share	transportation_access	inf
block5_transit_commute_share_preferred	transportation_access	inf
block5_tts_no_car_household_share	transportation_access	inf
block5_tts_transit_trip_share	transportation_access	inf
block4_mayoral_top_two_margin	mayoral_competitiveness	inf
block2_apartment_duplex_count	housing_form_density	inf
block2_apartment_5plus_storeys_count	housing_form_density	inf
block4_mayoral_winner_margin	mayoral_competitiveness	inf
block2_renter_share	housing_tenure	8100.055
block2_owner_share	housing_tenure	8033.196
block2_structural_type_total_dwellings	housing_form_density	284.495

Variable	Family	VIF
block2_condo_status_total_dwellings	housing_form_density	259.010
block4_federal_vote_fragmentation	federal_competitiveness	163.980
block4_effective_federal_parties_5pct	federal_competitiveness	154.937
block1_age_65_plus_share	age_structure	146.305
block1_age_18_34_share	age_structure	129.804
block4_provincial_vote_fragmentation	provincial_competitiveness	99.928
block4_effective_provincial_parties_5pct	provincial_competitiveness	89.222

Strongest Correlated Pairs

A	B	r	Flag
block5_no_car_household_share	block5_tts_no_car_household_share	1.000	near_duplicate_or_inverse
block5_transit_commute_share_preferred	block5_tts_transit_trip_share	1.000	near_duplicate_or_inverse
block4_mayoral_top_two_margin	block4_mayoral_winner_margin	1.000	near_duplicate_or_inverse
block2_renter_share	block2_owner_share	-1.000	near_duplicate_or_inverse
block2_structural_type_total_dwellings	block2_condo_status_total_dwellings	0.994	near_duplicate_or_inverse
block4_effective_federal_parties_5pct	block4_federal_vote_fragmentation	0.994	near_duplicate_or_inverse
block4_effective_provincial_parties_5pct	block4_provincial_vote_fragmentation	0.989	near_duplicate_or_inverse
block2_population_density_per_km2	block2_apartments_per_km2	0.971	near_duplicate_or_inverse
block4_effective_mayoral_candidates_5pct	block4_mayoral_vote_fragmentation	0.967	near_duplicate_or_inverse
block2_apartment_5plus_storeys_count	block2_apartment_total_count	0.960	near_duplicate_or_inverse
block2_same_address_1yr_share	block2_same_address_5yr_share	0.928	strong
block2_structural_type_total_dwellings	block2_apartment_total_count	0.924	strong
block2_apartment_total_count	block2_condo_status_total_dwellings	0.923	strong
block1_age_65_plus_share	block1_median_age	0.918	strong
block3_immigrant_share	block3_non_official_mother_tongue_share	0.916	strong
block3_non_citizen_share	block3_citizen_adult_share	-0.913	strong
block3_recent_immigrant_share	block3_non_citizen_share	0.899	strong
block3_immigrant_share	block3_visible_minority_share	0.889	strong
block2_apartment_5plus_storeys_count	block2_condominium_dwellings_count	0.876	strong
block2_apartment_5plus_storeys_count	block2_condo_status_total_dwellings	0.865	strong
block2_structural_type_total_dwellings	block2_apartment_5plus_storeys_count	0.865	strong
block2_apartment_share	block2_detached_share	-0.856	strong
block2_apartments_per_km2	block2_condos_per_km2	0.851	strong
block2_apartment_total_count	block2_condominium_dwellings_count	0.844	moderate_high
block1_age_18_34_share	block2_same_address_1yr_share	-0.827	moderate_high

Plain-Language Redundancy Summary

The most important high-correlation or inverse relationships are no car household share and TTS no car household share with correlation 1.000; transit commute share and TTS transit trip share with correlation 1.000; mayoral top two margin and mayoral winner margin with correlation 1.000; renter share and owner share with correlation -1.000; structural type total dwellings and condo status total dwellings with correlation 0.994; effective federal parties above five percent and federal vote fragmentation with correlation 0.994; effective provincial parties above five percent and provincial vote fragmentation with correlation 0.989; population density and apartments per square kilometre with correlation 0.971; effective mayoral candidates above five percent and mayoral vote fragmentation with correlation 0.967; apartment five plus storeys count and apartment total count with correlation 0.960. These are the relationships most likely to double-count the same social concept if they are fed into dimension reduction together.

Family-Level Redundancy

Family	Vars	Max	r	
mayoral_competitiveness	5	1.000		
transportation_access	5	1.000		
housing_tenure	2	1.000		
housing_form_density	14	0.994		
federal_competitiveness	4	0.994		
provincial_competitiveness	4	0.989		
residential_stability	2	0.928		
age_structure	4	0.918		
immigration_citizenship	7	0.916		
service_access	11	0.808		
service_contact	8	0.679		
household_class	4	0.677		

Interpretation

The diagnostics confirm Zack's warning: the modelling table contains variables that are not merely correlated but sometimes structurally tied. Tenure is the clearest example because renter and owner shares are inverse codings of the same concept. Housing form and density also form a dense cluster, especially because apartment counts, apartment shares, condo variables, density variables, and total dwelling counts partly track the same urban form.

VIF is stricter than pairwise correlation because it asks whether a variable can be reconstructed from all other predictors together. High VIF does not mean the variable is unimportant; it means the model cannot cleanly separate its unique contribution from nearby variables. That is why Step 3 should use family comparisons rather than deleting variables mechanically by VIF.

The practical conclusion is that the final cleaned model should choose representatives from correlated families, then test substitutions and connected-family combinations. That preserves the idea that A may work better with D than with C, which pairwise correlation alone cannot discover.

Step 3: Theory-Cleaned Supervised PLS and Variable Family Tournament

Step 3 uses the Step 1 VIP evidence and Step 2 redundancy diagnostics to build a more defensible supervised dimension-reduction model. The goal is not to pretend the predictors are independent. The goal is to keep a smaller set of representatives so no single social concept gets double-weighted in the PLS components.

Full vs Cleaned Model

Model	Predictors	Components	CV R2	CV RMSE	Train R2
Full unfiltered PLS	70	12	0.565	0.057	0.693
Theory-cleaned PLS	27	5	0.558	0.058	0.625

Selection Logic

The cleaned model was built with a family approach. Within each family, variables were judged using four pieces of evidence: common-sense interpretability, bivariate turnout association, full-model PLS VIP, and redundancy/VIF. Where variables were close substitutes, the workflow also ran one-family substitution tests and connected-family combination tests.

This means a variable could be excluded even if it had a high VIP when another selected variable represented the same concept more cleanly. Conversely, a variable could stay with moderate VIP if it gives a clearer theoretical reading and does not create severe redundancy.

The 5 selected components are latent variables created from the 27 selected original predictors. They are not 5 named raw variables. The model is interpreted by looking back at which original predictors have high VIP scores and strong loadings within those latent components.

Final Variables Kept

Variable	Family	VIP	Turnout corr	VIF	Reason
block1_age_65_plus_share	age_structure	1.219	0.182	146.305	Kept as a theory/common-sense representative and checked against diagnostics.
block1_age_18_34_share	age_structure	0.932	-0.136	129.804	Kept as a theory/common-sense representative and checked against diagnostics.
block4_federal_margin	federal_competitiveness	1.065	0.272	8.264	Kept as a theory/common-sense representative and checked against diagnostics.
block1_bachelors_or_higher_25_64_share	household_class	1.985	0.554	7.119	Kept as a theory/common-sense representative and checked against diagnostics.
block1_average_household_size	household_class	1.834	-0.485	26.655	Kept as a theory/common-sense representative and checked against diagnostics.
block1_low_income_lim_at_share	household_class	0.919	-0.199	10.691	Kept as a theory/common-sense representative and checked against diagnostics.
block2_apartment_share	housing_form_density	0.863	0.089	30.687	Kept as a theory/common-sense representative and checked against diagnostics.
block2_condo_share	housing_form_density	0.664	0.084	28.586	Kept as a theory/common-sense representative and checked against diagnostics.
block2_population_density_per_km2	housing_form_density	0.744	0.102	47.932	Kept as a theory/common-sense representative and checked against diagnostics.
block2_renter_share	housing_tenure	0.671	-0.081	8100.055	Kept as a theory/common-sense representative and checked against diagnostics.
block3_visible_minority_share	immigration_citizenship	2.286	-0.642	16.311	Kept as a theory/common-sense representative and checked against diagnostics.

Variable	Family	VIP	Turnout corr	VIF	Reason
block3_non_citizen_share	immigration_citizenship	1.534	-0.417	23.912	Kept as a theory/common-sense representative and checked against diagnostics.
block3_recent_immigrant_share	immigration_citizenship	1.227	-0.317	13.072	Kept as a theory/common-sense representative and checked against diagnostics.
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.907	-0.535	39.781	Kept as a theory/common-sense representative and checked against diagnostics.
block4_mayoral_top_two_margin	mayoral_competitiveness	1.151	0.288	inf	Kept as a theory/common-sense representative and checked against diagnostics.
block4_provincial_margin	provincial_competitiveness	0.802	0.184	4.606	Kept as a theory/common-sense representative and checked against diagnostics.
block2_same_address_5yr_share	residential_stability	0.644	-0.120	14.819	Kept as a theory/common-sense representative and checked against diagnostics.
block5_shelter_access_1200m	service_access	0.728	0.139	2.882	Kept as a theory/common-sense representative and checked against diagnostics.
block5_library_access_1200m	service_access	0.630	0.137	4.646	Kept as a theory/common-sense representative and checked against diagnostics.
block5_community_centre_access_1200m	service_access	0.495	-0.037	2.138	Kept as a theory/common-sense representative and checked against diagnostics.
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.933	0.157	3.371	Kept as a theory/common-sense representative and checked against diagnostics.
block5_social_housing_share	service_contact	0.747	-0.191	2.642	Kept as a theory/common-sense representative and checked against diagnostics.
block5_requests_311_per_1000	service_contact	0.765	0.140	10.169	Kept as a theory/common-sense representative and checked against diagnostics.
block5_development_applications_2021_2025_per_1000	service_contact	0.690	0.095	5.890	Kept as a theory/common-sense representative and checked against diagnostics.

Variable	Family	VIP	Turnout corr	VIF	Reason
block5_school_age_5_17_share	service_contact	0.882	-0.194	46.384	Kept as a theory/common-sense representative and checked against diagnostics.
block5_no_car_household_share	transportation_access	0.832	0.174	inf	Kept as a theory/common-sense representative and checked against diagnostics.
block5_transit_commute_share_preferred	transportation_access	0.538	-0.002	inf	Kept as a theory/common-sense representative and checked against diagnostics.

In plain language, the kept variables are age 65 plus share, age 18 to 34 share, bachelor or higher education share, average household size, low income share, renter share, same address five year share, apartment share, condo share, population density, visible minority share, non citizen share, recent immigrant share, effective mayoral candidates above five percent, mayoral top two margin, federal margin, provincial margin, no car household share, transit commute share, shelter access within 1200 metres, library access within 1200 metres, community centre access within 1200 metres, KSI collisions per 1000 residents, social housing share, 311 requests per 1000 residents, development applications per 1000 residents, school age children share.

Cleaned Model VIP Results

Variable	Family	VIP	Turnout corr	Direction
block3_visible_minority_share	immigration_citizenship	2.145	-0.642	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	1.847	0.554	higher turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.787	-0.535	lower turnout
block1_average_household_size	household_class	1.672	-0.485	lower turnout
block3_non_citizen_share	immigration_citizenship	1.433	-0.417	lower turnout
block3_recent_immigrant_share	immigration_citizenship	1.137	-0.317	lower turnout
block4_mayoral_top_two_margin	mayoral_competitiveness	1.084	0.288	lower turnout
block4_federal_margin	federal_competitiveness	0.967	0.272	higher turnout
block1_age_18_34_share	age_structure	0.833	-0.136	lower turnout
block5_school_age_5_17_share	service_contact	0.820	-0.194	higher turnout
block1_age_65_plus_share	age_structure	0.805	0.182	higher turnout
block1_low_income_lim_at_share	household_class	0.770	-0.199	lower turnout
block5_no_car_household_share	transportation_access	0.741	0.174	lower turnout
block5_shelter_access_1200m	service_access	0.716	0.139	lower turnout
block4_provincial_margin	provincial_competitiveness	0.675	0.184	higher turnout
block5_social_housing_share	service_contact	0.674	-0.191	lower turnout
block2_apartment_share	housing_form_density	0.669	0.089	higher turnout
block2_population_density_per_km2	housing_form_density	0.609	0.102	higher turnout
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.592	0.157	higher turnout
block2_same_address_5yr_share	residential_stability	0.575	-0.120	lower turnout

Remaining VIF After Cleaning

Variable	Family	Cleaned VIF
block1_average_household_size	household_class	17.521
block2_apartment_share	housing_form_density	14.575
block2_renter_share	housing_tenure	14.327
block1_age_18_34_share	age_structure	12.753
block3_non_citizen_share	immigration_citizenship	12.265
block5_school_age_5_17_share	service_contact	10.666
block2_condo_share	housing_form_density	10.022
block3_recent_immigrant_share	immigration_citizenship	9.402
block2_same_address_5yr_share	residential_stability	9.301
block3_visible_minority_share	immigration_citizenship	7.847
block1_low_income_lim_at_share	household_class	6.224
block1_age_65_plus_share	age_structure	5.412
block5_no_car_household_share	transportation_access	5.102
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	4.677
block1_bachelors_or_higher_25_64_share	household_class	3.890

The cleaned model removes the exact duplicate/inverse variables from the full universe, so the impossible VIFs mostly disappear. Some VIF values remain above 10 because the social geography itself is bundled: household size, age, tenure, apartment form, citizenship, and school-age composition still move together across Toronto CTs. That remaining collinearity should be handled by reading the PLS components and VIP scores, not by treating each coefficient as a standalone causal estimate.

The age variables are a good example of remaining conceptual correlation. Age 18 to 34 share and age 65 plus share are partly compositional, but they are not exact inverses. I kept both because they represent different turnout stories: younger-adult concentration and older-adult concentration. Still, the VIF table shows that age structure remains bundled with household size, residential stability, and school-age composition, so these should be interpreted as a family rather than isolated causal coefficients.

Reference Categories For The Cleaned Model

The cleaned model suggests five main reference categories for the final story: immigration and racialized geography, education and class resources, age and household composition, housing tenure and urban form, and electoral competitiveness. The most stable variables behind these categories are visible minority share, bachelor or higher education share, effective mayoral candidates above five percent, average household size, non citizen share, recent immigrant share, mayoral top two margin, federal margin, age 18 to 34 share, school age children share, age 65 plus share, low income share. These categories are a way to bridge the model-first workflow back to the kind of reference categories Zack was emphasizing.

Family Tournament Highlights

Family/test	Best candidate	CV R2	CV RMSE
age_structure	block1_age_18_34_share	0.559	0.058
age_structure+immigration_citizenship	block1_age_65_plus_share; block3_visible_minority_share	0.564	0.057
federal_competitiveness	block4_federal_margin	0.558	0.058
household_class	block1_bachelors_or_higher_25_64_share	0.558	0.058
housing_form_density	block2_apartment_share	0.559	0.058
housing_tenure	block2_renter_share	0.558	0.058
housing_tenure+housing_form_density	block2_renter_share; block2_apartment_share	0.559	0.058
immigration_citizenship	block3_visible_minority_share	0.566	0.057

Family/test	Best candidate	CV R2	CV RMSE
mayoral_competitiveness	block4_effective_mayoral_candidates_5pct	0.560	0.058
mayoral_competitiveness+federal_competitiveness+provincial_competitiveness	block4_effective_mayoral_candidates_5pct; block4_federal_margin; block4_provincial_margin	0.560	0.058
provincial_competitiveness	block4_effective_provincial_parties_5pct	0.562	0.058
residential_stability	block2_same_address_1yr_share	0.558	0.058
service_access	block5_shelter_count_1200m	0.557	0.058
service_contact	block5_ksi_collision_events_2021_2025_per_1000	0.569	0.057
transportation_access	block5_transit_commute_share_preferred	0.561	0.058
transportation_access+service_access+service_contact	block5_no_car_household_share; block5_shelter_count_1200m; block5_ksi_collision_events_2021_2025_per_1000	0.578	0.056

Variable-by-Variable Decisions

- block1_age_65_plus_share (age_structure): Kept. VIP 1.219, turnout correlation 0.182, VIF 146.305. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_age_18_34_share (age_structure): Kept. VIP 0.932, turnout correlation -0.136, VIF 129.804. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_median_age (age_structure): Not kept. VIP 1.268, turnout correlation 0.232, VIF 28.099. Excluded to keep the family from double-weighting a correlated concept.
- block1_age_35_64_share (age_structure): Not kept. VIP 0.708, turnout correlation 0.142, VIF 44.191. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_federal_margin (federal_competitiveness): Kept. VIP 1.065, turnout correlation 0.272, VIF 8.264. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_federal_party_count_5pct (federal_competitiveness): Not kept. VIP 0.912, turnout correlation 0.206, VIF 5.005. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_federal_vote_fragmentation (federal_competitiveness): Not kept. VIP 0.436, turnout correlation -0.057, VIF 163.980. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_effective_federal_parties_5pct (federal_competitiveness): Not kept. VIP 0.416, turnout correlation -0.029, VIF 154.937. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block1_bachelors_or_higher_25_64_share (household_class): Kept. VIP 1.985, turnout correlation 0.554, VIF 7.119. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_average_household_size (household_class): Kept. VIP 1.834, turnout correlation -0.485, VIF 26.655. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_low_income_lim_at_share (household_class): Kept. VIP 0.919, turnout correlation -0.199, VIF 10.691. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_unemployment_rate_share (household_class): Not kept. VIP 1.934, turnout correlation -0.532, VIF 3.556. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_share (housing_form_density): Kept. VIP 0.863, turnout correlation 0.089, VIF 30.687. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_condo_share (housing_form_density): Kept. VIP 0.664, turnout correlation 0.084, VIF 28.586. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_population_density_per_km2 (housing_form_density): Kept. VIP 0.744, turnout correlation 0.102, VIF 47.932. Kept as a theory/common-sense representative and checked against diagnostics.

- block2_semi_detached_share (housing_form_density): Not kept. VIP 0.570, turnout correlation 0.095, VIF 6.727. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_condos_per_km2 (housing_form_density): Not kept. VIP 0.579, turnout correlation 0.071, VIF 13.166. Excluded to keep the family from double-weighting a correlated concept.
- block2_detached_share (housing_form_density): Not kept. VIP 0.742, turnout correlation -0.047, VIF 25.246. Excluded to keep the family from double-weighting a correlated concept.
- block2_condominium_dwellings_count (housing_form_density): Not kept. VIP 0.571, turnout correlation 0.071, VIF 31.740. Excluded to keep the family from double-weighting a correlated concept.
- block2_apartments_per_km2 (housing_form_density): Not kept. VIP 0.673, turnout correlation 0.115, VIF 69.113. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_apartment_lt5_storeys_count (housing_form_density): Not kept. VIP 1.034, turnout correlation 0.275, VIF inf. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_duplex_count (housing_form_density): Not kept. VIP 0.866, turnout correlation -0.169, VIF inf. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_total_count (housing_form_density): Not kept. VIP 0.634, turnout correlation 0.096, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_structural_type_total_dwellings (housing_form_density): Not kept. VIP 0.543, turnout correlation 0.083, VIF 284.495. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_condo_status_total_dwellings (housing_form_density): Not kept. VIP 0.542, turnout correlation 0.083, VIF 259.010. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_apartment_5plus_storeys_count (housing_form_density): Not kept. VIP 0.566, turnout correlation 0.035, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_renter_share (housing_tenure): Kept. VIP 0.671, turnout correlation -0.081, VIF 8100.055. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_owner_share (housing_tenure): Not kept. VIP 0.666, turnout correlation 0.079, VIF 8033.196. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block3_visible_minority_share (immigration_citizenship): Kept. VIP 2.286, turnout correlation -0.642, VIF 16.311. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_non_citizen_share (immigration_citizenship): Kept. VIP 1.534, turnout correlation -0.417, VIF 23.912. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_recent_immigrant_share (immigration_citizenship): Kept. VIP 1.227, turnout correlation -0.317, VIF 13.072. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_immigrant_share (immigration_citizenship): Not kept. VIP 2.031, turnout correlation -0.573, VIF 34.659. Excluded to keep the family from double-weighting a correlated concept.
- block3_non_official_mother_tongue_share (immigration_citizenship): Not kept. VIP 1.751, turnout correlation -0.491, VIF 21.937. Excluded to keep the family from double-weighting a correlated concept.
- block3_citizen_adult_share (immigration_citizenship): Not kept. VIP 1.401, turnout correlation 0.353, VIF 9.215. Excluded to keep the family from double-weighting a correlated concept.
- block3_english_french_knowledge_share (immigration_citizenship): Not kept. VIP 0.969, turnout correlation 0.265, VIF 5.821. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_effective_mayoral_candidates_5pct (mayoral_competitiveness): Kept. VIP 1.907, turnout correlation -0.535, VIF 39.781. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_mayoral_top_two_margin (mayoral_competitiveness): Kept. VIP 1.151, turnout correlation 0.288, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_mayoral_vote_fragmentation (mayoral_competitiveness): Not kept. VIP 1.775, turnout correlation -0.495, VIF 55.498. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_mayoral_candidate_count_5pct (mayoral_competitiveness): Not kept. VIP 0.584, turnout correlation -0.107, VIF 1.952. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_mayoral_winner_margin (mayoral_competitiveness): Not kept. VIP 1.151, turnout correlation 0.288, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.

- block4_provincial_margin (provincial_competitiveness): Kept. VIP 0.802, turnout correlation 0.184, VIF 4.606. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_provincial_party_count_5pct (provincial_competitiveness): Not kept. VIP 0.688, turnout correlation -0.150, VIF 2.354. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_effective_provincial_parties_5pct (provincial_competitiveness): Not kept. VIP 0.529, turnout correlation -0.052, VIF 89.222. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_provincial_vote_fragmentation (provincial_competitiveness): Not kept. VIP 0.530, turnout correlation -0.061, VIF 99.928. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_same_address_5yr_share (residential_stability): Kept. VIP 0.644, turnout correlation -0.120, VIF 14.819. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_same_address_1yr_share (residential_stability): Not kept. VIP 0.748, turnout correlation -0.156, VIF 13.103. Excluded to keep the family from double-weighting a correlated concept.
- block5_shelter_access_1200m (service_access): Kept. VIP 0.728, turnout correlation 0.139, VIF 2.882. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_library_access_1200m (service_access): Kept. VIP 0.630, turnout correlation 0.137, VIF 4.646. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_community_centre_access_1200m (service_access): Kept. VIP 0.495, turnout correlation -0.037, VIF 2.138. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_library_count_1200m (service_access): Not kept. VIP 0.990, turnout correlation 0.246, VIF 4.853. Excluded to keep the family from double-weighting a correlated concept.
- block5_shelter_count_1200m (service_access): Not kept. VIP 0.818, turnout correlation 0.082, VIF 3.889. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_shelter_nearest_m (service_access): Not kept. VIP 0.718, turnout correlation -0.146, VIF 3.357. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_library_nearest_m (service_access): Not kept. VIP 0.680, turnout correlation -0.137, VIF 3.964. Excluded to keep the family from double-weighting a correlated concept.
- block5_park_count_1200m (service_access): Not kept. VIP 0.581, turnout correlation 0.108, VIF 1.800. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_community_centre_nearest_m (service_access): Not kept. VIP 0.558, turnout correlation 0.063, VIF 2.461. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_community_centre_count_1200m (service_access): Not kept. VIP 0.424, turnout correlation -0.040, VIF 1.644. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_park_nearest_m (service_access): Not kept. VIP 0.366, turnout correlation 0.066, VIF 1.264. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_ksi_collision_events_2021_2025_per_1000 (service_contact): Kept. VIP 0.933, turnout correlation 0.157, VIF 3.371. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_social_housing_share (service_contact): Kept. VIP 0.747, turnout correlation -0.191, VIF 2.642. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_requests_311_per_1000 (service_contact): Kept. VIP 0.765, turnout correlation 0.140, VIF 10.169. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_development_applications_2021_2025_per_1000 (service_contact): Kept. VIP 0.690, turnout correlation 0.095, VIF 5.890. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_school_age_5_17_share (service_contact): Kept. VIP 0.882, turnout correlation -0.194, VIF 46.384. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_requests_311_2023_2025_estimated_count (service_contact): Not kept. VIP 0.844, turnout correlation 0.168, VIF 11.940. Excluded to keep the family from double-weighting a correlated concept.
- block5_development_applications_2021_2025 (service_contact): Not kept. VIP 0.666, turnout correlation 0.120, VIF 3.590. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_ksi_collision_events_2021_2025 (service_contact): Not kept. VIP 0.618, turnout correlation -0.050, VIF 3.028. Excluded to keep the cleaned model interpretable and lower dimensional.

- block5_no_car_household_share (transportation_access): Kept. VIP 0.832, turnout correlation 0.174, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_transit_commute_share_preferred (transportation_access): Kept. VIP 0.538, turnout correlation -0.002, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_tts_overlap_area_m2 (transportation_access): Not kept. VIP 0.649, turnout correlation -0.102, VIF 6.638. Excluded to keep the family from double-weighting a correlated concept.
- block5_tts_no_car_household_share (transportation_access): Not kept. VIP 0.832, turnout correlation 0.174, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block5_tts_transit_trip_share (transportation_access): Not kept. VIP 0.538, turnout correlation -0.002, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.

Final Interpretation

The cleaned PLS model should be treated as the main supervised dimension-reduction candidate. It keeps turnout supervision through PLS, while reducing the most obvious double-counting problems that appear in the full model. If an inverse pair performs almost identically, the final choice should be explained as interpretive rather than purely predictive. For example, renter share and owner share can encode nearly the same tenure gradient with opposite signs; keeping renter share makes the urban turnout story easier to discuss without implying that owner share is empirically irrelevant.

The remaining ambiguous cases are valuable rather than embarrassing. They show where the data cannot clearly distinguish one representative from another, especially inside housing form, service access, and election-competitiveness families. Those should be reported as sensitivity areas, not hidden.

Step 4: Interaction Discovery After Cleaning

Step 4 tests interpretable interaction terms after the Step 3 family-cleaned model. The candidate pool is broader than the exact Step 3 model: it includes selected variables plus reasonable non-duplicate family alternatives. Variables removed as exact duplicates, inverse codings, or near-compositional redundancies are not reintroduced.

Baseline vs Interaction-Augmented PLS

Model	Predictors	Components	CV R2	CV RMSE	Train R2
Step 3 cleaned PLS	27	5	0.558	0.058	0.625
Step 4 interaction PLS	38	1	0.566	0.057	0.587

Screening Rule

Each candidate interaction was tested by adding the interaction plus its main effects if those main effects were not already in the cleaned model. The screen used cross-validated PLS performance and the interaction term's VIP. I treated an interaction as promising only when it did not worsen CV RMSE and had interaction VIP around 0.8 or higher.

Best Interaction Trials

Interaction	Families	CV R2	CV RMSE	VIP	Story
block1_low_income_lim_at_share__x__block5_requests_311_per_1000	household_class + service_contact	0.618	0.054	0.954	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_bachelors_or_higher_25_64_share__x__block5_requests_311_per_1000	household_class + service_contact	0.607	0.055	0.775	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.

Interaction	Families	CV R2	CV RMSE	VIP	Story
block5_ksi_collision_events_2021_2025_per_1000__x__block1_unemployment_rate_share	service_contact + household_class	0.596	0.055	0.808	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_low_income_lim_at_share__x__block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.595	0.055	0.645	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_average_household_size__x__block5_requests_311_per_1000	household_class + service_contact	0.593	0.055	0.711	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block5_ksi_collision_events_2021_2025_per_1000__x__block3_non_official_mother_tongue_share	service_contact + immigration_citizenship	0.591	0.056	1.086	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_bachelors_or_higher_25_64_share__x__block1_median_age	household_class + age_structure	0.584	0.056	0.774	Tests whether the class/household-size turnout pattern changes across age structure.
block5_requests_311_per_1000__x__block3_immigrant_share	service_contact + immigration_citizenship	0.582	0.056	0.651	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_low_income_lim_at_share__x__block1_age_35_64_share	household_class + age_structure	0.582	0.056	0.933	Tests whether the class/household-size turnout pattern changes across age structure.
block1_average_household_size__x__block5_ksi_collision_events_2021_2025_per_1000	household_class + service_contact	0.581	0.056	0.827	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block5_transit_commute_share_preferred__x__block5_requests_311_2023_2025_estimated_count	transportation_access + service_contact	0.580	0.056	0.593	Theory-screened cross-family interaction with interpretable main effects retained.
block5_development_applications_2021_2025_per_1000__x__block3_immigrant_share	service_contact + immigration_citizenship	0.580	0.056	0.613	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.

Interaction	Families	CV R2	CV RMSE	VIP	Story
block5_requests_311_per_1000__x__block3_non_official_mother_tongue_share	service_contact + immigration_citizenship	0.578	0.057	0.623	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_bachelors_or_higher_25_64_share__x__block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.577	0.057	0.618	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_bachelors_or_higher_25_64_share__x__block2_renter_share	household_class + housing_tenure	0.577	0.057	0.541	Theory-screened cross-family interaction with interpretable main effects retained.
block3_non_citizen_share__x__block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.577	0.057	0.403	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_age_65_plus_share__x__block1_average_household_size	age_structure + household_class	0.576	0.057	0.950	Tests whether the class/household-size turnout pattern changes across age structure.
block1_unemployment_rate_share__x__block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.576	0.057	0.629	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block3_visible_minority_share__x__block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.576	0.057	0.566	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_low_income_lim_at_share__x__block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.574	0.057	0.395	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_average_household_size__x__block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.573	0.057	0.282	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.

Interaction	Families	CV R2	CV RMSE	VIP	Story
block1_bachelors_or_higher_25_64_share__x__block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.572	0.057	0.605	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_age_65_plus_share__x__block1_bachelors_or_higher_25_64_share	age_structure + household_class	0.572	0.057	0.690	Tests whether the class/household-size turnout pattern changes across age structure.
block1_low_income_lim_at_share__x__block2_condo_share	household_class + housing_form_density	0.571	0.057	0.773	Theory-screened cross-family interaction with interpretable main effects retained.
block3_recent_immigrant_share__x__block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.571	0.057	0.352	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.

Interactions Retained In The Augmented PLS

Interaction	VIP	Turnout corr	Direction
block1_bachelors_or_higher_25_64_share__x__block4_effective_mayoral_candidates_5pct	0.863	0.240	higher turnout
block5_ksi_collision_events_2021_2025_per_1000__x__block3_non_official_mother_tongue_share	0.846	0.235	higher turnout
block1_low_income_lim_at_share__x__block1_age_35_64_share	0.798	-0.222	lower turnout
block5_ksi_collision_events_2021_2025_per_1000__x__block1_unemployment_rate_share	0.767	-0.213	lower turnout
block1_average_household_size__x__block1_age_35_64_share	0.753	0.210	higher turnout
block1_average_household_size__x__block5_ksi_collision_events_2021_2025_per_1000	0.725	-0.202	lower turnout
block1_low_income_lim_at_share__x__block5_requests_311_per_1000	0.704	0.196	higher turnout
block1_age_65_plus_share__x__block1_average_household_size	0.572	-0.159	lower turnout

Interpretation Of Retained Interactions

- bachelor or higher education share with effective mayoral candidates above five percent: this asks whether education or class resources change how local electoral fragmentation relates to turnout. It is useful for a story about whether complex local contests are easier to navigate in higher-resource places.
- KSI collisions per 1000 residents with non official mother tongue share: this asks whether service-contact geography has a different meaning in immigrant, language, or citizenship contexts. It may point to places where municipal service contact and civic participation are linked unevenly.
- low income share with age 35 to 64 share: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.

- KSI collisions per 1000 residents with unemployment rate: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- average household size with age 35 to 64 share: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.
- average household size with KSI collisions per 1000 residents: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- low income share with 311 requests per 1000 residents: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- age 65 plus share with average household size: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.

Reference Categories For Interactions

The interaction results mainly point to four conditional reference categories: class and service contact, education and local electoral fragmentation, language or immigrant geography and service contact, and age structure interacting with class or household size. These should not all become final hypotheses automatically. They are candidate story lines that can be checked against maps, plots, and substantive plausibility.

Remaining VIF In Interaction Model

Variable	Family	VIF
block1_age_18_34_share	age_structure	118.541
block1_age_65_plus_share	age_structure	103.655
block5_school_age_5_17_share	service_contact	45.526
block1_age_35_64_share	age_structure	38.511
block1_average_household_size	household_class	26.825
block5_ksi_collision_events_2021_2025_per_1000__x__block1_unemployment_rate_share	other	16.688
block1_low_income_lim_at_share__x__block5_requests_311_per_1000	other	16.204
block2_apartment_share	housing_form_density	15.919
block5_ksi_collision_events_2021_2025_per_1000__x__block3_non_official_mother_tongue_share	other	15.433
block2_renter_share	housing_tenure	15.194
block3_non_citizen_share	immigration_citizenship	14.031
block1_average_household_size__x__block5_ksi_collision_events_2021_2025_per_1000	other	11.071
block2_condo_share	housing_form_density	10.652
block2_same_address_5yr_share	residential_stability	10.401
block3_recent_immigrant_share	immigration_citizenship	10.269

Interpretation

Interactions should be used as story devices only when they describe a plausible conditional relationship. The strongest candidates here are not arbitrary products; they mainly ask whether the turnout penalty associated with social composition becomes sharper in places with particular housing form, electoral fragmentation, or service/contact contexts. If an interaction improves prediction only trivially, it should still be kept out of the final narrative unless it clarifies a substantive mechanism.

Step 5A: Sparse PLS Comparison

Sparse PLS keeps the PLS idea but forces each component to use only a limited number of variables. This implementation is a transparent sparse-PLS approximation: within each component, it keeps only the largest predictor weights before deflation. This is useful here because ordinary PLS can spread importance across correlated variables, while sparse PLS asks which variables still matter when each latent component must be simpler.

Model Comparison

Model	Predictors	Components	CV R2	CV RMSE
Full unfiltered PLS	70	12	0.565	0.057
Step 3 cleaned PLS	27	5	0.558	0.058
Sparse PLS	27	5	0.560	0.058

Best sparse setting: keep 15 variables per component.

Top Sparse PLS Variables

Variable	Family	VIP	In Sparse Weights	Components
block3_visible_minority_share	immigration_citizenship	2.217	True	3
block1_bachelors_or_higher_25_64_share	household_class	1.921	True	2
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.861	True	3
block1_average_household_size	household_class	1.731	True	3
block3_non_citizen_share	immigration_citizenship	1.444	True	3
block4_mayoral_top_two_margin	mayoral_competitiveness	1.211	True	2
block3_recent_immigrant_share	immigration_citizenship	1.146	True	5
block4_federal_margin	federal_competitiveness	0.956	True	3
block1_age_65_plus_share	age_structure	0.877	True	3
block5_school_age_5_17_share	service_contact	0.823	True	4
block1_age_18_34_share	age_structure	0.753	True	2
block5_no_car_household_share	transportation_access	0.737	True	5
block1_low_income_lim_at_share	household_class	0.721	True	3
block4_provincial_margin	provincial_competitiveness	0.719	True	3
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.711	True	3
block5_social_housing_share	service_contact	0.689	True	2
block2_apartment_share	housing_form_density	0.583	True	3
block2_condo_share	housing_form_density	0.540	True	3
block5_shelter_access_1200m	service_access	0.529	True	2
block2_population_density_per_km2	housing_form_density	0.449	True	2

Interpretation

Sparse PLS is mainly a readability test. If the same variables remain important under sparsity, that strengthens the story. If performance collapses, it suggests turnout is being explained by a wider bundled social geography rather than a small handful of variables.

Sparse PLS did not introduce new raw variables beyond the cleaned predictor set. Instead, it reweighted the same cleaned variables under a sparsity constraint. The variables that remain strongest under this stricter setup are visible minority share, bachelor or higher education share, effective mayoral candidates above five percent, average household size, non citizen share, mayoral top two margin, recent immigrant share, federal margin, age 65 plus share, school age children share, age 18 to 34 share, no car household share. This supports the same reference categories as the cleaned PLS model: immigration and racialized geography, education and class resources, household composition, electoral competitiveness, age structure, and selected service-contact context.

Step 5B: Supervised PCA Comparison

Supervised PCA first screens variables by their relationship with turnout, then runs ordinary PCA on the screened variable set. It is less directly supervised than PLS, but easier to explain: turnout chooses the variable set; PCA summarizes the selected predictors.

Model Comparison

Model	Predictors/Screened Vars	Components	CV R2	CV RMSE
Full unfiltered PLS	70	12	0.565	0.057
Step 3 cleaned PLS	27	5	0.558	0.058
Supervised PCA	15	8	0.540	0.059

Best screen: corr_ge_0.15.

Largest PCA Loadings In Best Screen

Variable	Max	loading	
block5_ksi_collision_events_2021_2025_per_1000	0.633		
block1_age_65_plus_share	0.630		
block4_mayoral_top_two_margin	0.554		
block1_bachelors_or_higher_25_64_share	0.513		
block5_social_housing_share	0.498		
block4_provincial_margin	0.488		
block1_low_income_lim_at_share	0.434		
block4_effective_mayoral_candidates_5pct	0.430		
block5_no_car_household_share	0.403		
block3_visible_minority_share	0.402		
block4_federal_margin	0.401		
block3_non_citizen_share	0.396		
block3_recent_immigrant_share	0.389		
block5_school_age_5_17_share	0.358		
block1_average_household_size	0.355		

Interpretation

Supervised PCA is a useful comparison because it asks whether a simple turnout-screened latent structure can compete with PLS. If it performs similarly, the final story can lean more on interpretable screened dimensions. If it underperforms, PLS remains the stronger supervised reduction method.

The best supervised PCA screen retained these plain-language variables: visible minority share, bachelor or higher education share, effective mayoral candidates above five percent, average household size, non citizen share, recent immigrant share, mayoral top two margin, federal margin, low income share, school age children share, social housing share, provincial margin, age 65 plus share, no car household share, KSI collisions per 1000 residents. This model is useful as a reference-category check because it independently returns a compact set around racialized and citizenship geography, education and class, household structure, electoral competitiveness, age structure, social housing, carlessness, and KSI collisions. Because its predictive performance is weaker than PLS, I would use it as supporting evidence rather than the main model.

Step 5C: Elastic Net Robustness Check

Elastic Net is not a latent-variable method. It is included here as a robustness check on the interaction-augmented candidate set: if a variable or interaction is important in PLS/VIP and also survives penalized regression, that is stronger evidence that it belongs in the final story.

Best Elastic Net Setting

- Alpha: 0.03
- L1 ratio: 0.5
- Mean selected variables across folds: 27.000
- CV R2 on scaled turnout: 0.617
- CV RMSE on scaled turnout: 0.619

Selected Variables

Variable	Family	Abs scaled coef	Direction
block3_visible_minority_share	immigration_citizenship	0.291	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	0.202	higher turnout
block1_age_18_34_share	age_structure	0.180	lower turnout
block2_apartment_share	housing_form_density	0.125	higher turnout
block1_bachelors_or_higher_25_64_share__x__block4_effective_mayoral_candidates_5pct	other	0.123	higher turnout
block1_average_household_size	household_class	0.114	lower turnout
block2_population_density_per_km2	housing_form_density	0.104	higher turnout
block1_low_income_lim_at_share__x__block5_requests_311_per_1000	other	0.088	higher turnout
block3_non_citizen_share	immigration_citizenship	0.086	lower turnout
block4_federal_margin	federal_competitiveness	0.077	higher turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	0.069	lower turnout
block3_non_official_mother_tongue_share	immigration_citizenship	0.068	lower turnout
block5_ksi_collision_events_2021_2025_per_1000__x__block1_unemployment_rate_share	other	0.066	lower turnout
block1_low_income_lim_at_share	household_class	0.059	lower turnout
block5_community_centre_access_1200m	service_access	0.052	lower turnout
block1_average_household_size__x__block5_ksi_collision_events_2021_2025_per_1000	other	0.049	lower turnout
block2_condo_share	housing_form_density	0.043	higher turnout
block1_age_65_plus_share__x__block1_average_household_size	other	0.040	lower turnout
block1_low_income_lim_at_share__x__block1_age_35_64_share	other	0.036	lower turnout
block1_average_household_size__x__block1_age_35_64_share	other	0.032	higher turnout
block2_renter_share	housing_tenure	0.032	lower turnout
block5_shelter_access_1200m	service_access	0.026	lower turnout
block5_library_access_1200m	service_access	0.010	lower turnout
block5_requests_311_per_1000	service_contact	0.008	higher turnout

Variable	Family	Abs scaled coef	Direction
block5_social_housing_share	service_contact	0.004	higher turnout

Agreement With Interaction-Augmented PLS

Top interaction-augmented PLS variables also selected by Elastic Net:

- block1_average_household_size
- block1_bachelors_or_higher_25_64_share
- block1_bachelors_or_higher_25_64_share__x__block4_effective_mayoral_candidates_5pct
- block3_non_citizen_share
- block3_non_official_mother_tongue_share
- block3_visible_minority_share
- block4_effective_mayoral_candidates_5pct
- block4_federal_margin
- block4_mayoral_top_two_margin

Interpretation

Elastic Net should not replace PLS for this project because it does not create latent components. Its value is diagnostic. Variables and interactions that survive both methods are sturdy candidates for the final narrative. Terms that are high-VIP but not selected by Elastic Net may still matter as part of a latent geography, but they are less convincing as standalone predictors.

The strongest Elastic Net terms in plain language are visible minority share, bachelor or higher education share, age 18 to 34 share, apartment share, bachelor or higher education share with effective mayoral candidates above five percent, average household size, population density, low income share with 311 requests per 1000 residents, non citizen share, federal margin, effective mayoral candidates above five percent, non official mother tongue share. As a robustness check, this supports reference categories around racialized geography, education, young adults, apartment and density context, local electoral fragmentation, household size, low income and 311 service contact, citizenship, and federal competitiveness.